Arya Zadebrahimi

AI Specialist

aryazady@gmail.com +989199242710 Tehran, Iran

Online Resume

Professional Summary

Bioinformatics graduate with practical experience in AI/ML/DL, data science, and software development. Trained, fine-tuned and evaluated models in generative AI, LLMs, GNNs, NLP, image processing, offline reinforcement learning, and imitation learning, with familiarity in RAG and diffusion models. Proficient in PyTorch Geometric, Hugging Face, and Ollama, with a strong foundation in algorithms and statistics.

Education

Sharif University of Technology — Master of Science • GPA 18.2
Bioinformatics — Sep 2021 – Jul 2024

Islamic Azad University — Bachelor of Science • GPA 15.58
Software Engineering — Sep 2015 – May 2021

Experience

Iran University of Medical Sciences — Tehran, Iran
Intern — Aug 2022
Completed hands-on training in molecular biology techniques, including cell lysis and DNA extraction, PCR-based detection of specific DNA sequences, cell culture, and microscopic cell counting.

Nobaar — Tehran, Iran
Technical Lead — Jun 2018 – Dec 2018
Developed two applications for customers and company drivers, and led the design and architecture of the application APIs, collaborating with the backend development team on their implementation.

Mibarim — Tehran, Iran
Android Developer — Oct 2017 – Mar 2018
Developed two applications supporting ridesharing services and a separate app for comparing transportation options across ridesharing platforms.

Projects

Financial Market Prediction
A transformer-based model for predicting financial market based on a window of previous behavior.

Breast Cancer Classification
A Hierarchical vision transformer model to classify positive and negative cases.

Persian Document NER Tagging
A bidirectional LSTM-CRF model for NER tagging a Persian document

Brain MRI Semantic Segmentation
A DDPM-based semantic segmentation model for brain MRI

Skills

AI / Bininformatics
Training, Fine-Tuning, Evaluation, Natural Language Processing, Large Language Models, Image Processing, Diffusion Models, Generative AI, Mathematical Modeling, Constrained Optimization, Sequence Alignment, scRNA Expression Analysis, FASTA/FASTQ File Processing, Molecular Docking

Coding
Python, Pytorch, R / Limma • Bioconductor, Julia, ReactJS, NodeJS, Android, Java, TCL

Databases
MongoDB, MySQL, SQLite, NCBI, UniProt, PDB, ChEMBL, PubChem, BiGG

Tools & Software
Hugging Face, Ollama, HISAT2, Samtools, BFCtools, Beagle, SHAPEIT, Guassian, AutoDock Vina, CellProfiler, VMD

Laboratory Techniques
DNA extraction, PCR, Cell culture